Dang Truong Nguyen

Software Engineer Intern

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EDUCATION

VNU - University of Science, Ho Chi Minh City, Vietnam **2023 - 2027**
Bachelor of Information Technology - Major in Software Engineering **Current GPA: 8.92/10**
Relevant Coursework: Web Development, Database Systems, Software Engineering, Computer Networks and Cloud Computing, CI/CD and DevOps.

- Encouragement Prize, Vietnam Datathon - DataStorm 2025.
- Top 11, NASA Space Apps Challenge 2025 - Ho Chi Minh City.
- University's Study Encouragement Scholarship – Semester 2 - 2023

Thang Long High School for the Gifted, Da Lat City, Vietnam **2020 - 2023**
Physics Major

- 1st Prize, Lam Dong Provincial Excellent Student Competition (2023).
- 2nd Prize, Lam Dong Provincial Excellent Student Competition (2022).
- Reserve Member, Lam Dong Provincial Team for the National Excellent Student Competition (2022).

SKILLS

- **Languages:** C++, JavaScript, Python, SQL.
- **Frontend:** React, NextJs, TailwindCSS.
- **APIs & Serving:** FastAPI, Flask, ExpressJS, NodeJs.
- **Datastores:** MongoDB, Mongoose, MySQL, PostgreSQL.
- **Tools:** Git, CI with Jenkins/Github Action, Docker, AWS.

CERTIFICATIONS

- IELTS 6.5 - 2023
- AWS Academy Graduate - Cloud Architecting - Training Badge
- AWS Certified Solutions Architect - Associate (AWS SAA-C03 763/1000) - 2026

PROJECTS

Task Management System with AWS Serverless Architecture Task Management - Demo
A full-stack task management application leveraging AWS Lambda, API Gateway, DynamoDB, S3 and CloudFront for a scalable serverless solution.
My Role (Team of 3)

- Configured VPC, IAM roles, and security groups to ensure traffic only travel through API Gateway, the DynamoDB is isolated from the public internet.
- Designed and implemented a serverless backend using AWS Lambda functions and API Gateway for RESTful API endpoints.
- Deploy a static frontend on AWS S3 with CloudFront using Origin Access Identity for secure content delivery.

Features

- Task creation, editing, deletion, and categorization with real-time updates.
- Serverless architecture for automatic scaling and reduced operational overhead.

- Monitoring and logging with AWS CloudWatch for performance insights and troubleshooting.
- Cost control created with AWS Budgets with limit set to 0.01 USD per month to prevent unexpected costs.

Tech Stack

- Frontend: HTML, CSS, JavaScript.
- Backend: AWS Lambda, API Gateway.
- Database: DynamoDB.

DataStorm Encouragement Prize, Vietnam Datathon - DataStorm 2025

An end-to-end retail analytics and SKU-level demand forecasting platform for FMCG, integrating time-series modeling with interactive dashboards.

My Role (Team of 5)

- Built a full-stack analytics system with data ingestion pipelines, scalable RESTful APIs, and an interactive dashboard using Next.js and Tailwind CSS.
- Optimized performance by implementing Redis caching and query optimization to reduce API response time.
- Integrated LLM-based suggestions and deployed services across Vercel (frontend), Render (backend), and AWS EC2 (AI workloads).

Features

- Interactive sales analytics dashboard with geo-based and channel-level drill-down capabilities.
- 7-day demand forecasting module for inventory optimization and automated stock alerting.
- Price elasticity modeling and promotion impact analysis for data-driven pricing decisions.
- Multi-country store benchmarking through advanced geospatial visualization.

Tech Stack

- Frontend: NextJs, TailwindCSS, Zustand, React's Support Lib for UI (shadcn/ui, recharts).
- Backend: FastAPI, SQLAlchemy.
- Database: PostgreSQL, Redis.
- Hosting: AWS E2C Instance for AI server, PaaS Render for Backend server and Vercel for Frontend.
- LLM Calling: Google Gemini
- Machine Learning & Forecasting: Pandas, Scikit-learn, XGBoost.

AirForce Top 11, NASA Space Apps Challenge 2025 – Ho Chi Minh City

A air quality monitoring and 7-day AQI forecasting platform integrating satellite, ground-station, and meteorological data streams.

My Role (Team of 6)

- Developed an interactive coordinate-based UI with real-time geospatial visualizations.
- Integrated frontend with Node.js/Python backend via RESTful APIs and deployed services on Vercel and Render.

Features

- Real-time data ingestion pipeline integrating satellite and ground-based environmental monitoring sources.
- Deep learning-powered 7-day AQI forecasting engine for short-term air quality prediction.
- Automated AQI alerting system aligned with WHO and EPA threshold standards.
- Temporal and geospatial visualization for comprehensive air quality monitoring and trend analysis.

Tech Stack

- Frontend: ReactJs, TailwindCSS, React's Support Lib for UI (Leaflet, AOS, . . .)
- Backend: FlaskAPI, NodeJs, Express.
- Hosting: PaaS Render for Backend server and Vercel for Frontend.
- LLM Calling: Google Gemini.
- Machine Learning & Forecasting: PyTorch, Pandas, NumPy, LSTM.